

BM40A0801

Machine Vision and Digital Image Analysis

Image-Level Micro Gesture Classification

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Team Members

Student 1: Maham Asif 002489137

Student 2: Sehar Gohar 002453312

Responsibilities

- **Student 1:** Model implementation and training
- **Student 2:** Evaluation, interpretation and report writing.

Abstract

This project focuses on the task of image-level micro-gesture classification using deep learning techniques. The dataset consists of 32 different gesture classes, which makes the classification problem challenging due to the similarity between gestures.

A Convolutional Neural Network based on the ResNet-50 architecture was used for this task. The model was initialized with pre-trained weights from ImageNet and modified to classify 32 classes. The dataset was divided into training and validation sets, and appropriate preprocessing techniques such as resizing and normalization were applied.

The model was trained and evaluated on the validation dataset, achieving an accuracy of 87.47%. A confusion matrix and per-class accuracy were used to further analyze the performance of the model.

The results show that the model performs well for this classification task. However, some classes remain challenging due to similarities in gestures. Overall, the proposed approach provides a reliable solution for micro-gesture classification.

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1 Introduction

Machine vision is an important field in computer science that focuses on enabling machines to understand and interpret visual data. It is widely used in applications such as surveillance systems, human-computer interaction, and medical imaging.

In this project, the task is to perform micro-gesture classification using a dataset that contains 32 different gesture classes. Micro-gestures are small and subtle human movements, which makes them difficult to detect and classify accurately.

The main objective of this project is to design and implement a deep learning model that can accurately classify these gestures into their respective classes. The dataset was divided into training and validation sets, where the validation set was used to evaluate the performance of the model.

The work in this project was divided between the team members based on different stages of the implementation.

One team member was responsible for the model development and training phase. This included data loading, preprocessing, defining the ResNet-50 architecture, and training the model using appropriate optimization techniques.

The second team member was responsible for the evaluation and analysis of the model. This included designing the evaluation pipeline, computing validation accuracy, generating and interpreting the confusion matrix, analyzing class-wise performance, and documenting the results in the report.

2 Literature Review

Traditional image classification methods mainly relied on hand-crafted features such as edges, textures, and shapes. Various feature extraction techniques were used to extract features from images. However, these methods depend heavily on manual feature design and are not very effective for complex datasets [1].

In recent years, deep learning approaches have significantly improved image classification performance. Convolutional Neural Networks (CNNs) are widely used because they can automatically learn important features directly from the data [2].

Among different CNN architectures, ResNet (Residual Network) is one of the most popular models. ResNet introduces skip connections, which help to solve the vanishing gradient problem and allow the training of very deep networks. This makes the model more efficient and improves its performance compared to traditional CNN architectures [3].

Another important concept is transfer learning. In this approach, models that are already trained on large datasets such as ImageNet are reused for new tasks. This reduces training time and improves performance, especially when the available dataset is limited [4].

These modern approaches have been successfully applied in many computer vision tasks, including gesture recognition. Therefore, in this project, a ResNet-based model with transfer learning was used for micro-gesture classification.

3 Method Description

In this project, a deep learning-based approach was used to solve the micro-gesture classification problem. The main model used for this task is ResNet-50, which is a convolutional neural network known for its strong performance in image classification tasks.

The ResNet-50 model was initialized with pre-trained weights from the ImageNet dataset. Transfer learning was used in this project to improve performance and reduce training time. Since the model was originally trained on a large dataset, it is able to extract useful features from images. The final fully connected layer of the model was modified to output 32 classes, which correspond to the gesture categories in the dataset.

The data preprocessing and model training phase included resizing all images to 224×224 pixels and normalizing them using standard mean and standard deviation values. The dataset was loaded using the ImageFolder class and divided into training and validation sets, where 80% of the data was used for training and 20% for validation. The model was trained using the CrossEntropyLoss function and the Adam optimizer for a fixed number of epochs.

After the training phase, the evaluation process was carried out to measure the performance of the model. This included computing the validation accuracy, generating a confusion matrix, and analyzing the classification results across different classes. These evaluation steps help in understanding the strengths and limitations of the model.

The trained model was also saved and used for prediction on new input images. A prediction function was implemented to classify unseen images and provide confidence scores for the predicted classes.

The ResNet-50 model was chosen for this project because it provides a good balance between accuracy and computational efficiency. Its use of residual connections allows it to train deeper networks effectively, making it suitable for complex classification tasks such as micro-gesture recognition.

4 Experiments and Results

The model was trained on the training dataset and evaluated using the validation dataset. The performance of the model was measured after training.

The model achieved a validation accuracy of 87.47%, which indicates that it can correctly classify most of the micro-gesture images.

To further analyze the performance, a confusion matrix was generated, as shown in Figure 1. The confusion matrix provides a detailed comparison between predicted and actual classes.

The confusion matrix shows that most values are concentrated along the diagonal, which means that the model correctly predicts the majority of the classes. Some classes have strong diagonal values, indicating high accuracy.

However, there are some off-diagonal values, which represent misclassifications. These errors mainly occur between classes with similar visual patterns, making them harder to distinguish.

Per-class accuracy was also calculated to evaluate the performance for each class. The results show that while many classes have high accuracy, a few classes perform lower due to similarity between gestures.

A manual test was also performed by selecting one sample image from each class. The model correctly classified most of these images, showing that it performs well on unseen

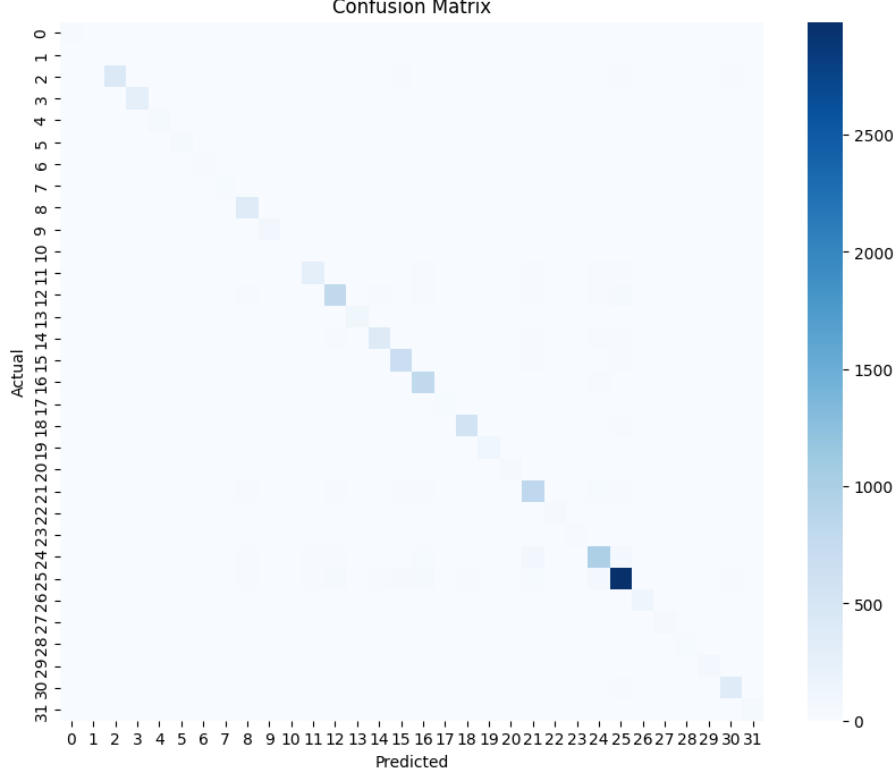


Figure 1: Confusion matrix of the classification results

samples.

Overall, the results indicate that the model performs well for micro-gesture classification, although there is still room for improvement in distinguishing similar classes.

5 Discussion

The results obtained in this project show that the proposed model is able to perform the classification task effectively. The validation accuracy of 87.47% indicates that the model generalizes well to unseen data and is capable of learning meaningful features from the dataset.

From the confusion matrix and per-class accuracy analysis, it can be observed that most classes are classified correctly with high accuracy. However, some classes show lower performance. This is likely due to the similarity between certain micro-gestures, which makes it difficult for the model to distinguish between them. In addition, class imbalance or a smaller number of training samples for some classes may also affect the performance.

The use of a pre-trained ResNet-50 model played an important role in achieving good results. Transfer learning allowed the model to utilize previously learned features, which reduced training time and improved accuracy. Without transfer learning, training a deep model from scratch would require more data and computational resources.

Despite the good performance, there are still some limitations in this approach. The model relies only on RGB image data and does not consider temporal or motion information, which could be useful for micro-gesture recognition. Since the dataset images are extracted from videos, incorporating temporal information could improve the classification results.

For future work, several improvements can be considered. Data augmentation techniques could be applied to increase the diversity of the training data and reduce overfitting. Additionally, using multimodal approaches, such as combining RGB images with skeleton data, may provide more discriminative features and improve performance. Exploring more advanced architectures or fine-tuning strategies could also lead to better results.

Overall, the model provides a reliable and efficient solution for the given classification task, and the results demonstrate the effectiveness of deep learning methods for micro-gesture recognition.

References

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